



ELECTRONIC CIRCUIT ANALYSIS

QUESTION BANK

MODULE- I:

SHORT ANSWER TYPE QUESTIONS:

1. List out the types of amplifiers based on coupling method.

Ans: Depending upon the method of coupling one stage to the other, there are RC coupled, Transformer coupled and direct coupled amplifier.

- 1. RC coupled Amplifier:** A multistage amplifier circuit that is coupled to the next stage using resistor and capacitor (RC) combination can be called as a RC coupled amplifier.
- 2. Transformer coupled Amplifier:** A multistage amplifier circuit that is coupled to the next stage, with the help of a transformer can be called as a Transformer coupled amplifier.
- 3. Direct-coupled amplifier:** A multistage amplifier circuit that is coupled to the next stage directly and can be called as Directly coupled amplifier.

2. List out the types of amplifiers based on transistor configuration.

Ans: Depending upon the type of transistor configuration, there are CE, CB and CC amplifiers.

- 1. CE amplifiers:** The CE amplifier circuit that is formed using a CE configured transistor combination is called as CE amplifier.
- 2. CB Amplifier:** The amplifier circuit that is formed using a CB configured transistor combination is called as CB amplifier.
- 3. CC Amplifier:** The amplifier circuit that is formed using a CC configured transistor combination is called as CC amplifier.

3. What is distortion in amplifiers. List out the different types of distortion.

Ans: Distortion in Amplifier basically implies the variation in the waveform received at the output with respect to the applied input. The unwanted alterations generated during amplification is known as distortion.



Distortion is a disturbance or change in shape of output signal. Distortion is of three types:

1. Harmonic distortion
2. Frequency distortion
3. Delay distortion or phase shift distortion

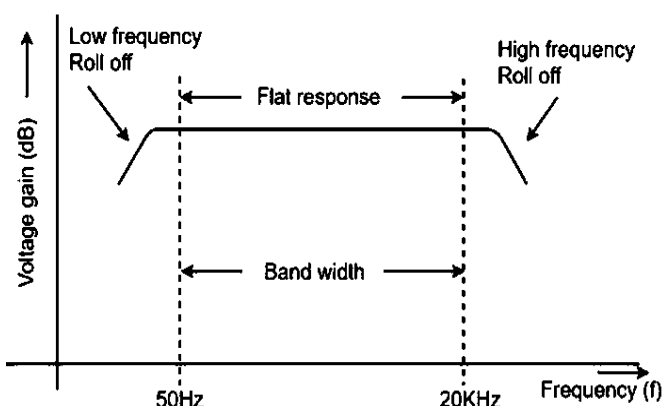
4. What is the main purpose of coupling device.

Ans: The basic purpose of a coupling device are

- **Transferring the AC Signal:** It successfully transfers the AC (alternating current) signal from the output of one amplification stage to the input of the next stage. This ensures the signal continues to be strengthened through the successive stages.
- **Blocking the DC Component:** It blocks the DC (direct current) voltage from passing from the output of one stage to the input of the next stage. This isolates the DC biasing conditions of each individual stage, preventing them from interfering with one another and distorting the circuit's stability.

5. Draw the frequency response of RC coupled amplifier and write the major drawback of RC coupled amplifier.

Ans: Frequency response:



The main drawback of RC coupled amplifier is when the 2 stages are coupled to make a multistage amplifier, the high output impedance of one stage comes in parallel with the low input impedance of next stage. Hence, effective load resistance is decreased. As this problem can be overcome by a transformer coupled amplifier.



6. List out the features of Darlington pair.

Ans: Darlington pair amplifier is combination of two common-collector configuration i.e., first stage amplifier will be CC configuration and second stage amplifier is also CC configuration.

Features of Darlington pair amplifier:

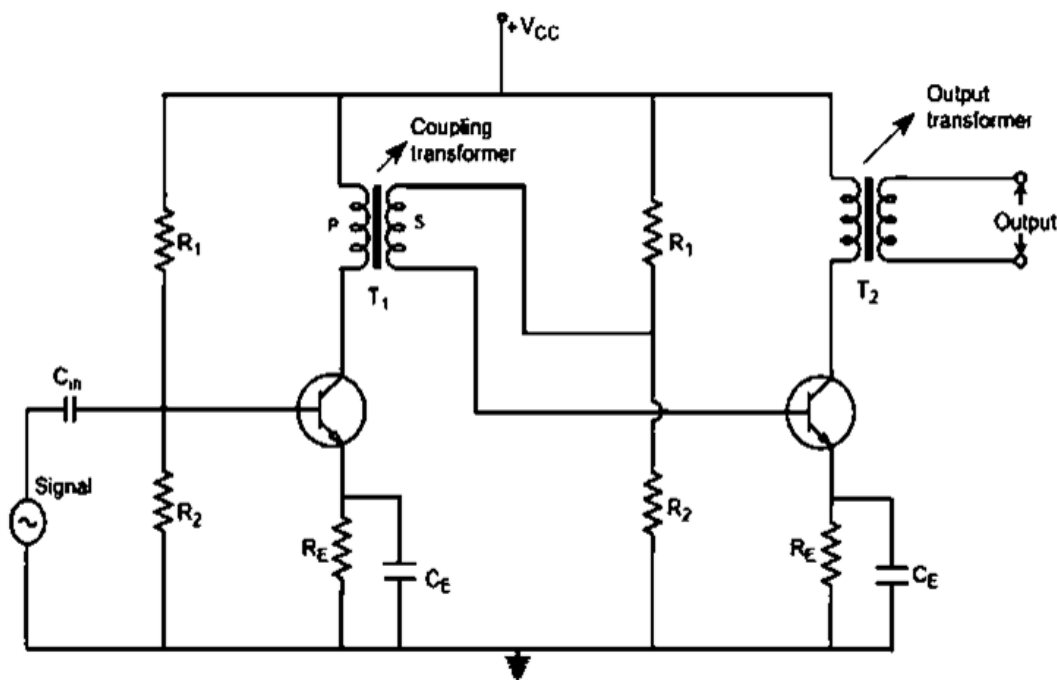
- The main characteristics of darlington pair configuration is that it provides higher current gain than the individual stages.
- This configuration occupies lesser space than two individual transistors due to the usage of common collector.
- It provides excellent buffering between a high impedance source and a low impedance load.

LONG ANSWER TYPE QUESTIONS:

1. With neat sketch, explain the principle of operation of transformer coupled amplifier and draw the Frequency response.

Ans: In a transformer-coupled amplifier, the stages of amplifier are coupled using a transformer.

CIRCUIT DIAGRAM:



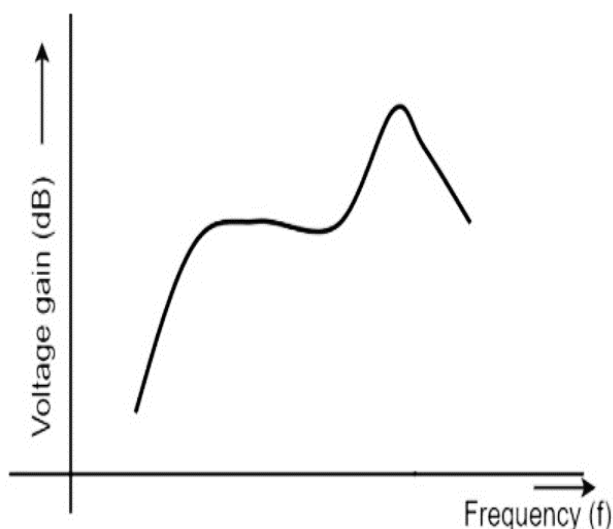


Operation:

- When an AC signal is applied to the input of the base of the first transistor then it gets amplified by the transistor and appears at the collector to which the primary of the transformer is connected.
- The transformer which is used as a coupling device in this circuit has the property of impedance changing, which means the low resistance of a stage (or load) can be reflected as a high load resistance to the previous stage.
- Hence the voltage at the primary is transferred according to the turns ratio of the secondary winding of the transformer.
- This transformer coupling provides good impedance matching between the stages of amplifier. The transformer coupled amplifier is generally used for power amplification.

FREQUENCY RESPONSE:

The figure below shows the frequency response of a transformer coupled amplifier. The gain of the amplifier is constant only for a small range of frequencies. The output voltage is equal to the collector current multiplied by the reactance of primary.





2. Compare different coupling schemes in amplifiers.

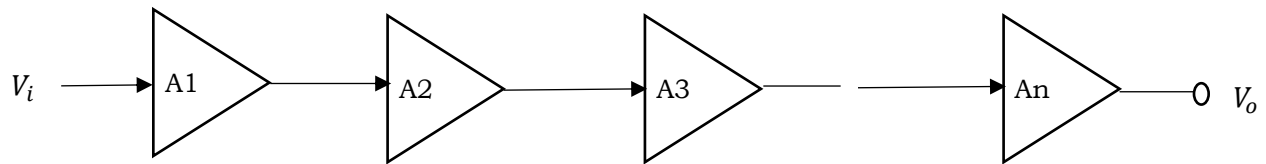
Ans:

S.No.	RC Coupling	Transformer Coupling	Direct Coupling
1	Resistors & Capacitors are used for coupling.	Impedance matching transformer is used.	No additional components are used for coupling.
2	It blocks DC Voltage.	It blocks DC Voltage.	It does not block DC voltage.
3	Frequency response is flat at middle frequencies.	Frequency response is non-uniform.	Frequency response is flat at middle frequencies.
4	Impedance matching is not possible.	Impedance matching is possible.	Impedance matching is not possible.
5	No voltage drift at the output.	No Voltage drift at the output.	Voltage drift occurs at the Output.
6	No DC amplification.	No DC amplification.	It performs DC amplification.
7	Mainly used in small signal audio amplifiers.	Used mainly where impedance matching is the main criteria.	Used in laboratories, where slow varying parameters are required.

3. Show the effect of lower cutoff and upper cutoff frequencies of multistage amplifiers.

Ans: Effect of Cutoff frequencies of multistage amplifiers:

As multistage amplifiers are classified into Interacting and Non-interacting amplifiers
Let us consider N non-interacting amplifiers which are connected in cascade having f_{L_1} & f_{H_1} as individual lower and higher cutoff frequencies.



for one stage,

$$A_{V_{L1}} = \frac{A_{V_{M1}}}{\sqrt{1 + \left(\frac{f_{L1}}{f}\right)^2}}$$

But for n-stages,

$$A_{V_L} = |A_{V_{L1}}| |A_{V_{L2}}| \dots |A_{V_{LN}}|$$

$$A_{V_L} = |A_{V_{L1}}|^N$$

Similarly,

$$A_{V_M} = |A_{V_{M1}}|^N$$

$$|A_{V_{L1}}|^N = \frac{|A_{V_{M1}}|^N}{\left[\sqrt{1 + \left(\frac{f_{L1}}{f}\right)^2} \right]^N}$$

$$A_{V_L} = \frac{A_{V_M}}{\left[\sqrt{1 + \left(\frac{f_{L1}}{f}\right)^2} \right]^N}$$

$$\text{At } f = f_L : A_{V_L} = \frac{A_{V_M}}{\sqrt{2}}$$

$$\frac{A_{V_M}}{\sqrt{2}} = \frac{A_{V_M}}{\left[\sqrt{1 + \left(\frac{f_{L1}}{f_L}\right)^2} \right]^N}$$



$$\left[\sqrt{1 + \left(\frac{f_{L_1}}{f_L} \right)^2} \right]^N = 2$$

$$1 + \left(\frac{f_{L_1}}{f_L} \right)^2 = 2^{1/N}$$

$$\left(\frac{f_{L_1}}{f_L} \right)^2 = 2^{1/N} - 1$$

$$\frac{f_{L_1}}{f_L} = \sqrt{2^{1/N} - 1}$$

$$f_L = \frac{f_{L_1}}{\sqrt{2^{1/N} - 1}}$$

The above expression indicates lower 3dB frequency (or) lower cutoff frequency (f_{L_1}).

Effect of higher cutoff frequency:

$$A_{V_{H_1}} = \frac{A_{V_{M_1}}}{\sqrt{1 + \left(\frac{f}{f_{H_1}} \right)^2}}$$

$$A_{V_H} = |A_{V_{H_1}}| \cdot |A_{V_{H_1}}| \dots \dots \dots N\text{-times}$$

$$A_{V_H} = |A_{V_{H_1}}|^N$$

$$\text{Similarly } A_{V_M} = |A_{V_{M_1}}|^N$$

$$|A_{V_{H_1}}|^N = \frac{|A_{V_{M_1}}|^N}{\left[\sqrt{1 + \left(\frac{f}{f_{H_1}} \right)^2} \right]^N}$$

$$A_{V_H} = \frac{A_{V_M}}{\left[1 + \left(\frac{f}{f_{H_1}} \right)^2 \right]^{N/2}}$$



$$\text{At } f = f_H \quad A_{V_H} = \frac{A_{V_M}}{\sqrt{2}}$$

$$\frac{A_{V_M}}{\sqrt{2}} = \frac{A_{V_M}}{\left[\sqrt{1 + \left(\frac{f_H}{f_{H_1}} \right)^2} \right]^N}$$

$$\left[1 + \left(\frac{f_H}{f_{H_1}} \right)^2 \right]^N = 2$$

$$1 + \left(\frac{f_H}{f_{H_1}} \right)^2 = 2^{1/N}$$

$$\frac{f_H}{f_{H_1}} = \sqrt{2^{1/N} - 1}$$

$$\boxed{f_H = f_{H_1} \sqrt{2^{1/N} - 1}}$$

- As $\sqrt{2^{1/N} - 1} < 1$ for a multistage amplifiers increases f_L and f_H decreases.

B.W = $f_H \sqrt{2^{1/N} - 1}$, so Bandwidth decreases.

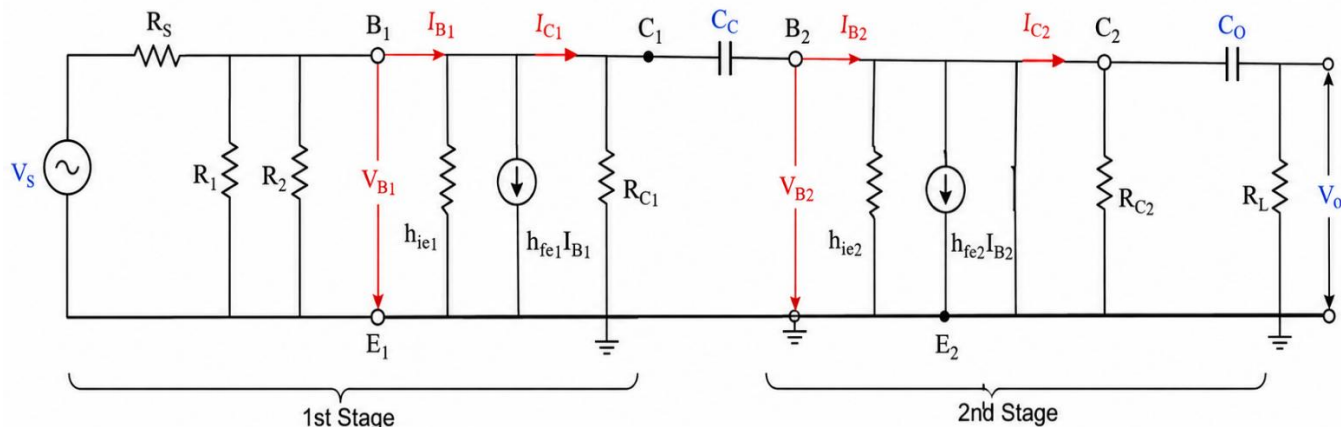
- For a multistage amplifier lower cutoff frequency increases, higher cutoff frequency decreases, resulting in decreasing bandwidth.



4. Draw the H- parameter model for Cascaded RC coupled amplifier and derive the expressions for Input impedance and Output admittance.

Ans:

RC-Coupled Cascade Amplifier (Approximate h-Parameter Model)



INPUT IMPEDANCE:

Input Impedance of the First Stage

The input impedance is defined as

$$Z_{i1} = \frac{\text{Input Voltage}}{\text{Input Current}} = \frac{V_{b1}}{I_{b1}}$$

Applying Ohm's law to Loop 1,

$$V_{b1} = I_{b1} h_{ie1}$$

Substituting,

$$Z_{i1} = \frac{I_{b1} h_{ie1}}{I_{b1}} = h_{ie1}$$

Hence,

$$Z_{i1} = h_{ie1}$$



Input Impedance of the Second Stage

Similarly,

$$Z_{i2} = \frac{V_{b2}}{I_{b2}}$$

Applying Ohm's law,

$$V_{b2} = I_{b2} h_{ie2}$$

Therefore,

$$Z_{i2} = \frac{I_{b2} h_{ie2}}{I_{b2}} = h_{ie2}$$

Hence,

$$Z_{i2} = h_{ie2}$$

The input source is connected only to the first stage. Therefore,

$$Z_i = R_1 \parallel R_2 \parallel h_{ie1}$$

If the biasing resistors are neglected (or replaced by their equivalent R_B), then

$$Z_i \approx h_{ie1}$$

Overall input impedance of the cascade amplifier:

The overall input impedance is determined only by the **first stage**, because the signal source is connected only to that stage. Thus,

$$Z_i = R_1 \parallel R_2 \parallel h_{ie1} \approx h_{ie1}$$

OUTPUT ADMITTANCE (Y_o)

For 1st Stage

The output admittance of the first stage is defined as

$$Y_{o1} = \frac{\text{Output Current of 1st Stage}}{\text{Output Voltage of 1st Stage}} = \frac{I_{C1}}{V_{C1}} \rightarrow (1)$$



From Loop 2,

$$V_{C1} = I_{C1}R_{C1} \rightarrow (2)$$

Substituting Eq. (2) into Eq. (1),

$$Y_{o1} = \frac{I_{C1}}{I_{C1}R_{C1}} = \frac{1}{R_{C1}}$$

$$Y_{o1} = \frac{1}{R_{C1}}$$

For 2nd Stage

The output admittance of the second stage is

$$Y_{o2} = \frac{\text{Output Current of 2nd Stage}}{\text{Output Voltage of 2nd Stage}} = \frac{I_L}{V_o} \rightarrow (3)$$

From Loop 4,

$$V_o = I_L R_L \rightarrow (4)$$

Substituting Eq. (4) into Eq. (3),

$$Y_{o2} = \frac{I_L}{I_L R_L} = \frac{1}{R_L}$$

$$Y_{o2} = \frac{1}{R_L}$$

Overall Output Admittance

Since the output resistances are connected in parallel,

$$Y_o = Y_{o1} + Y_{o2}$$

Therefore,

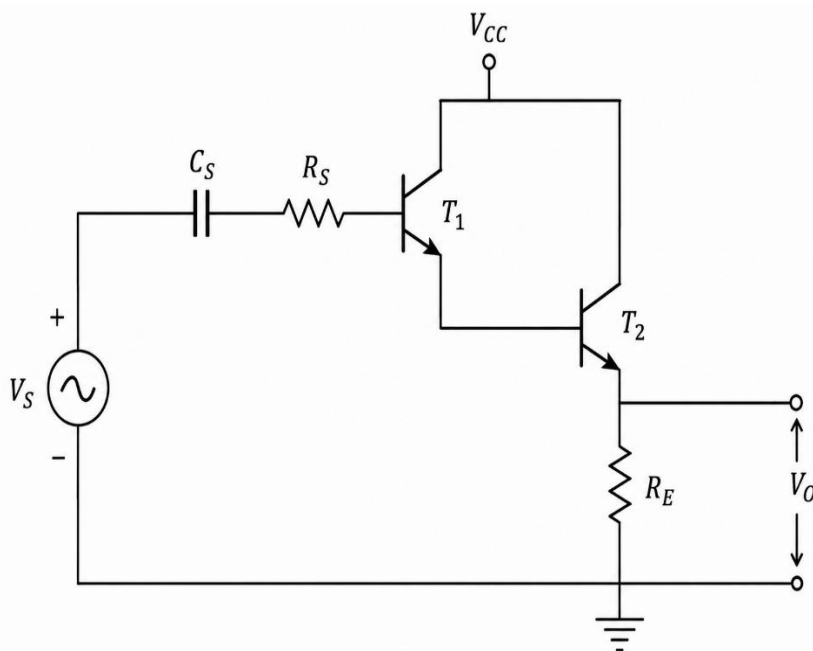
$$Y_o = \frac{1}{R_{C1}} + \frac{1}{R_L}$$



5. With a neat sketch, explain the working of Darlington pair amplifier and Draw the simplified h- parameter model.

Ans:

CIRCUIT DIAGRAM:

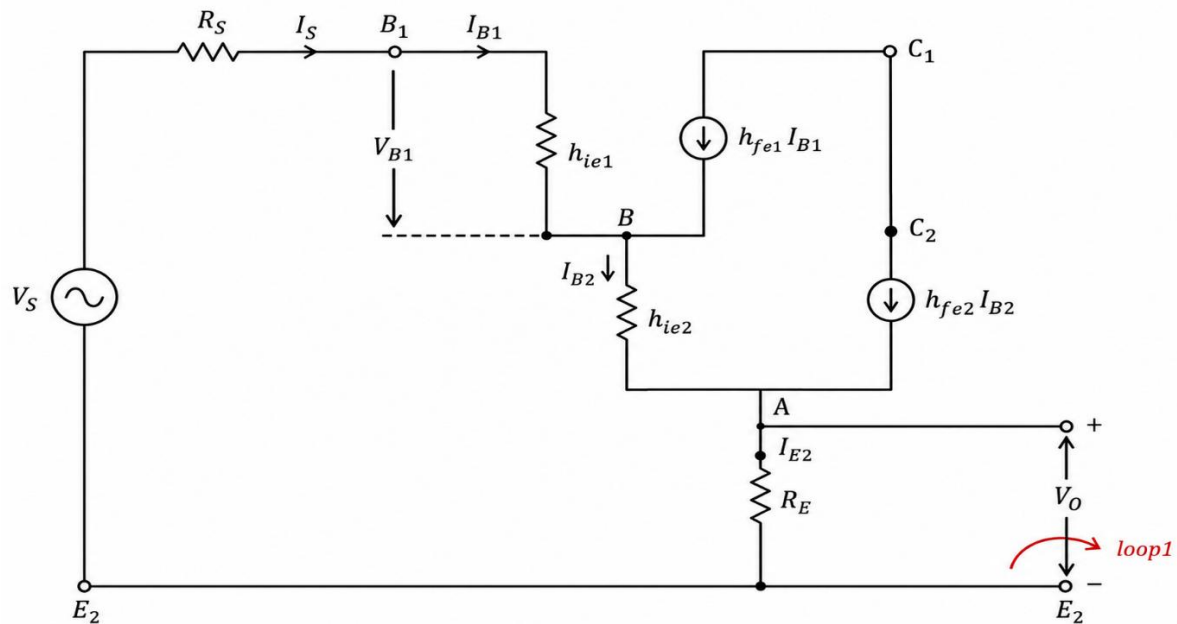


Construction:

- Two transistors T_1 and T_2 of the same type (both NPN or both PNP) are used.
- The collector terminals of both transistors are connected together and tied to the supply voltage V_{CC} .
- The emitter of the first transistor (T_1) is connected directly to the base of the second transistor (T_2).
- The input signal is applied to the base of T_1 through a coupling capacitor (C_S) and source resistance (R_S).
- The output is taken from the emitter of T_2 .
- An emitter resistor (R_E) is connected between the emitter of T_2 and ground to provide thermal stability and develop the output voltage.



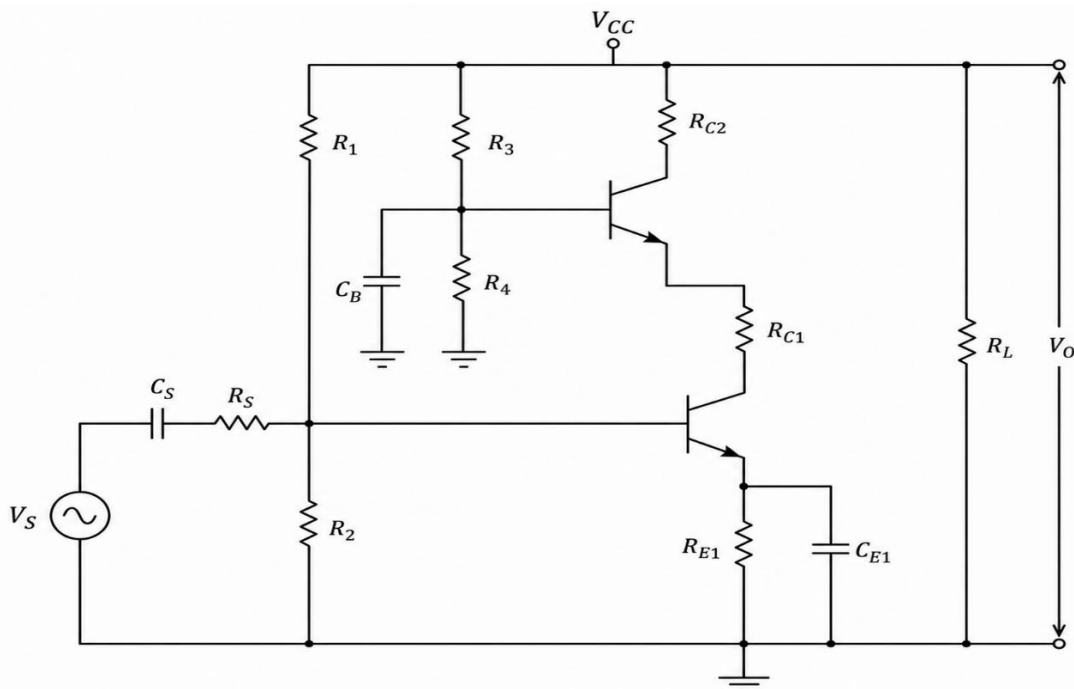
SIMPLIFIED H- PARAMETER MODEL:

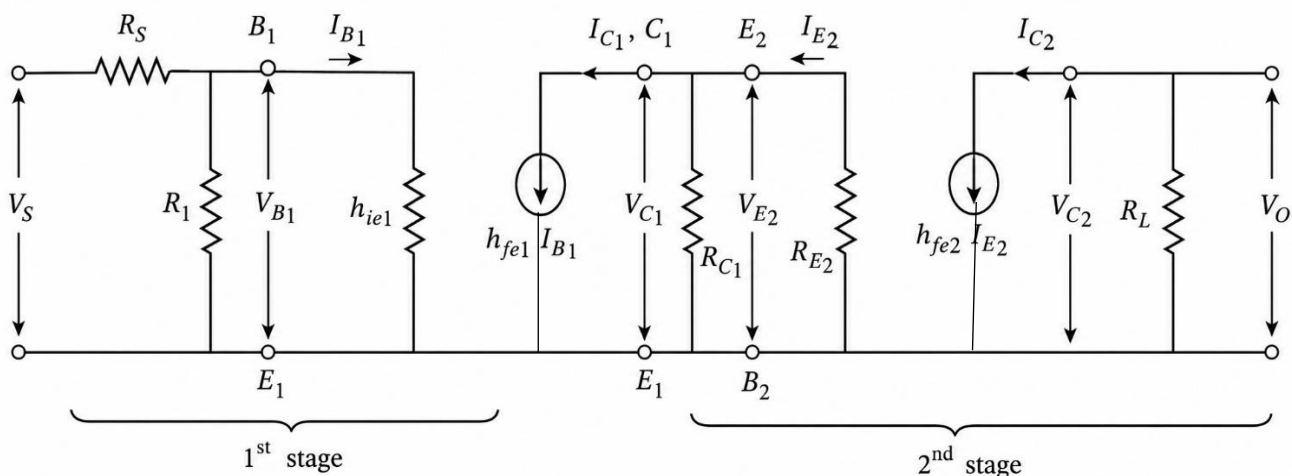


6. Draw the circuit diagram of Cascode amplifier and derive the expressions for current gain A_I and voltage gain A_V .

Ans:

CIRCUIT DIAGRAM:





CURRENT GAIN (A_I):

For the First Stage

The current gain of the first stage is defined as

$$A_{I1} = \frac{\text{Output current of 1st stage}}{\text{Input current of 1st stage}}$$

$$A_{I1} = \frac{I_{C1}}{I_{B1}} \rightarrow (1)$$

If fig (3), if we apply KCL at the terminals C_1 & E_1 ,

$$I_{C1} = h_{fe1} I_{B1} \rightarrow (2)$$

then

$$A_{I1} = \frac{h_{fe1} I_{B1}}{I_{B1}}$$

$$A_{I1} = h_{fe1} \rightarrow (3)$$

For the Second Stage

The current gain of the second stage is

$$A_{I2} = \frac{\text{Output current of 2nd stage}}{\text{Input current of 2nd stage}}$$



$$A_{I2} = \frac{I_L}{I_{E2}} \rightarrow (4)$$

If fig (3), if we apply KCL at terminals C_2 and B_2 ,

$$I_{C2} = h_{fe2}I_{E2} \rightarrow (5)$$

and

$$I_L = -I_{C2}$$

$$I_L = -h_{fe2}I_{E2}$$

$$A_{I2} = \frac{-h_{fe2}I_{E2}}{I_{E2}}$$

$$A_{I2} = -h_{fe2} \rightarrow (6)$$

W.K.T. the conversion form of

$$-h_{fb} = \frac{h_{fe}}{1 + h_{fe}}$$

Therefore,

$$A_{I2} = \frac{h_{fe2}}{1 + h_{fe2}} \rightarrow (7)$$

The overall current gain is given by

$$A_I = (A_{I1})(A_{I2})$$

$$A_I = (h_{fe1}) \left(\frac{h_{fe2}}{1 + h_{fe2}} \right)$$



VOLTAGE GAIN (A_V):

Voltage gain for 1st stage

$$A_{V1} = \frac{\text{Output voltage of 1st stage}}{\text{Input voltage of 1st stage}}$$

$$A_{V1} = \frac{V_{C1}}{V_{B1}} \rightarrow (1)$$

W.K.T.

$$V_{B1} = I_{B1}h_{ie1}$$

and

$$V_{C1} = -I_{C1}R_{C1}$$

$$[\because I_{C1} = I_{B1}h_{fe1}]$$

Therefore,

$$A_{V1} = \frac{-I_{C1}R_{C1}}{I_{B1}h_{ie1}}$$

$$A_{V1} = \frac{-I_{B1}h_{fe1}R_{C1}}{I_{B1}h_{ie1}}$$

$$\boxed{A_{V1} = -\frac{h_{fe1}R_{C1}}{h_{ie1}}} \rightarrow (2)$$



For 2nd Stage

$$A_{V2} = \frac{\text{Output voltage of 2nd stage}}{\text{Input voltage of 2nd stage}} = \frac{V_O}{V_{E2}} \rightarrow (3)$$

From Fig. (4), from loop (5), we get

$$V_{E2} = I_{E2}h_{ib2} \rightarrow (4)$$

From loop (3),

$$V_O = I_L R_L = -I_{C2} R_L = -h_{fb2} I_{E2} R_L \rightarrow (5)$$

Substituting equations (4) and (5) into (3),

$$\begin{aligned} A_{V2} &= \frac{V_O}{V_{E2}} \\ &= \frac{-h_{fb2} I_{E2} R_L}{I_{E2} h_{ib2}} \end{aligned}$$

$$A_{V2} = -\frac{h_{fb2} R_L}{h_{ib2}} \rightarrow (6)$$

Overall Voltage Gain

The overall voltage gain is given by

$$A_V = (A_{V1})(A_{V2})$$

$$A_V = \left(-\frac{h_{fe1} R_{C1}}{h_{ie1}} \right) \left(-\frac{h_{fb2} R_L}{h_{ib2}} \right)$$

Part A – Short Answer Questions

1. Define feedback in an amplifier. State the difference between positive and negative feedback.

Ans: Definition of Feedback:

Feedback in an amplifier is the process of taking a part of the output signal and feeding it back to the input of the amplifier.

Difference between Positive and Negative Feedback:

Positive Feedback	Negative Feedback
The feedback signal is in phase with the input signal.	The feedback signal is 180° out of phase with the input signal.
It increases the overall gain of the amplifier.	It decreases the overall gain of the amplifier.
It reduces stability and may cause oscillations.	It improves stability and reduces distortion and noise.
Used in oscillators.	Used in amplifiers to improve performance

2. Classify feedback amplifiers. Name the four basic feedback configurations.

Ans

Classification of Feedback Amplifiers

Feedback amplifiers are classified based on:

1. The quantity sampled at the output (Voltage or Current).
2. The method of mixing the feedback signal at the input (Series or Shunt).

Four Basic Feedback Configurations

1. Voltage-Series (Series-Shunt) Feedback
 - Output voltage is sampled.
 - Feedback is applied in series with the input.
2. Voltage-Shunt (Shunt-Shunt) Feedback
 - Output voltage is sampled.
 - Feedback is applied in shunt (parallel) with the input.

3. Current-Series (Series-Series) Feedback
 - Output current is sampled.
 - Feedback is applied in series with the input.
4. Current-Shunt (Shunt-Series) Feedback
 - Output current is sampled.
 - Feedback is applied in shunt (parallel) with the input.

3. List any six general characteristics (advantages) of negative feedback amplifiers.

Ans:

General Characteristics (Advantages) of Negative Feedback Amplifiers

Negative feedback amplifiers have the following characteristics:

1. Improved stability of gain – The amplifier gain becomes less sensitive to component variations and temperature changes.
2. Reduced distortion – Nonlinear distortion is minimized, improving signal quality.
3. Reduced noise – Unwanted noise in the amplifier is reduced.
4. Increased bandwidth – The frequency range over which the amplifier operates effectively is increased.
5. Improved input and output impedances – Input and output resistances can be adjusted to desired values depending on the feedback configuration.
6. Better overall stability – The amplifier becomes more stable and less likely to oscillate.

4. Explain the effect of negative feedback on amplifier gain, bandwidth, and distortion.

Effect of Negative Feedback on Amplifier Performance

1. Gain

Negative feedback reduces the overall gain of the amplifier.

However, the gain becomes more stable and less affected by temperature and component variations.

2. Bandwidth

Negative feedback increases the bandwidth of the amplifier.

The amplifier can amplify signals over a wider range of frequencies.

3. Distortion

Negative feedback reduces nonlinear distortion in the output signal.

This improves the fidelity and quality of the amplified signal.

5. Differentiate between Voltage-Series Feedback and Voltage-Shunt Feedback.

Voltage-Series Feedback	Voltage-Shunt Feedback
Output voltage is sampled.	Output voltage is sampled.
Feedback is connected in series with the input.	Feedback is connected in shunt (parallel) with the input.
Input resistance increases.	Input resistance decreases.
Output resistance decreases.	Output resistance decreases.
Used in voltage amplifiers.	Used in transresistance (current-to-voltage) amplifiers.

6. Differentiate between Current-Series Feedback and Current-Shunt Feedback.

Current-Series Feedback	Current-Shunt Feedback
Output current is sampled.	Output current is sampled.
Feedback is connected in series with the input.	Feedback is connected in shunt (parallel) with the input.
Input resistance increases.	Input resistance decreases.
Output resistance increases.	Output resistance increases.
Used in transconductance (voltage-to-current) amplifiers.	Used in current amplifiers.

Part B – Long Answer Questions (6 Marks Each)

1. Explain the concept of feedback in amplifiers. Classify feedback amplifiers and compare positive and negative feedback with suitable diagrams and applications.

Answer

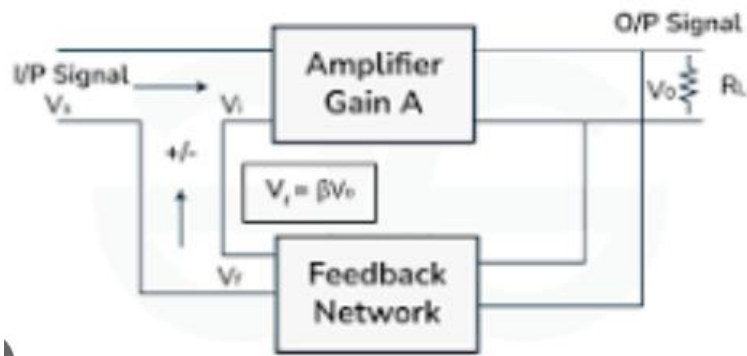
Concept of Feedback in Amplifiers

Feedback is the process of taking a portion of the output signal of an amplifier and feeding it back to the input.

If the feedback signal is added to the input, it is called positive feedback. If it is subtracted from the input, it is called negative feedback.

Block Diagram of a Feedback Amplifier

The feedback network takes a fraction (β) of the output and feeds it back to the input.



Classification of Feedback Amplifiers

Feedback amplifiers are classified based on:

1. Quantity sampled at the output

Voltage sampling

Current sampling

2. Method of mixing at the input

Series mixing

Shunt (parallel) mixing

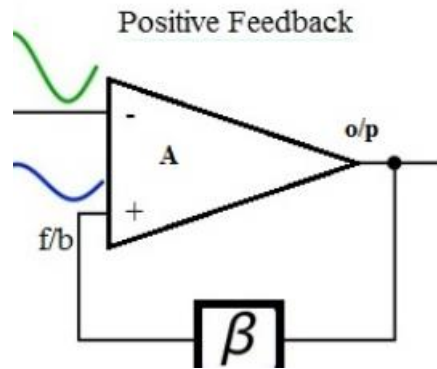
Four Basic Feedback Configurations

Configuration	Output Sampled	Input Mixing
Voltage-Series (Series-Shunt)	Voltage	Series
Voltage-Shunt (Shunt-Shunt)	Voltage	Shunt
Current-Series (Series-Series)	Current	Series
Current-Shunt (Shunt-Series)	Current	Shunt

Positive Feedback

In positive feedback, the feedback signal is in phase with the input signal and is added to it.

Block Diagram



Characteristics

- Increases amplifier gain.
- Reduces stability.
- May produce oscillations.
- Increases distortion and noise.

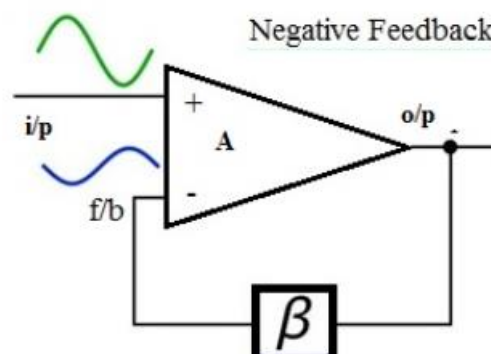
Applications

- Oscillators
- Signal generators
- Regenerative circuits

Negative Feedback

In negative feedback, the feedback signal is 180° out of phase with the input and is subtracted from it.

Block Diagram



Characteristics

- Reduces gain.

- Improves stability.
- Reduces distortion and noise.
- Increases bandwidth.
- Improves frequency response.

Applications

- Audio amplifiers
- Operational amplifier circuits
- Communication systems
- Instrumentation amplifiers

Comparison of Positive and Negative Feedback

Positive Feedback	Negative Feedback
Feedback is added to the input.	Feedback is subtracted from the input.
Increases gain.	Decreases gain.
Reduces stability.	Improves stability.
Increases distortion and noise.	Reduces distortion and noise.
Narrow bandwidth.	Wide bandwidth.
Used in oscillators.	Used in amplifiers.

Advantages of Negative Feedback

- Stable gain
- Reduced distortion
- Reduced noise
- Increased bandwidth
- Improved frequency response
- Better input and output impedance characteristics

2. Describe the general characteristics of negative feedback amplifiers

Answer:

Negative feedback amplifiers are amplifiers in which a portion of the output signal is fed back to the input in opposite phase. This feedback improves the overall performance and stability of the amplifier.

General Characteristics of Negative Feedback Amplifiers

Reduced Gain

The overall voltage gain decreases compared to the open-loop gain.

Gain becomes more stable and predictable.

Improved Stability

Gain is less affected by changes in temperature, transistor parameters, and power supply variations.

Reduced Distortion

Nonlinear distortion is significantly reduced, resulting in a cleaner output signal.

Reduced Noise

Internal noise generated by the amplifier is minimized, improving signal quality.

Increased Bandwidth

The frequency range over which the amplifier operates effectively is increased.

Improved Frequency Response

The amplifier provides a flatter and more uniform response across a wider frequency range.

Control of Input and Output Impedance

Negative feedback can increase or decrease input and output impedance depending on the feedback configuration used.

Better Linearity

The output signal follows the input signal more accurately, reducing nonlinear effects.

Improved Overall Performance

The amplifier becomes more reliable, accurate, and suitable for practical applications.

Advantages (Summary)

Stable gain

Lower distortion

Lower noise

Wider bandwidth

Better frequency response

Improved linearity

Controlled input/output impedance

3. Explain the Voltage-Series (Series-Shunt) Feedback amplifier with a neat block diagram. Discuss its working, advantages, and effect on input and output resistance.

Answer:

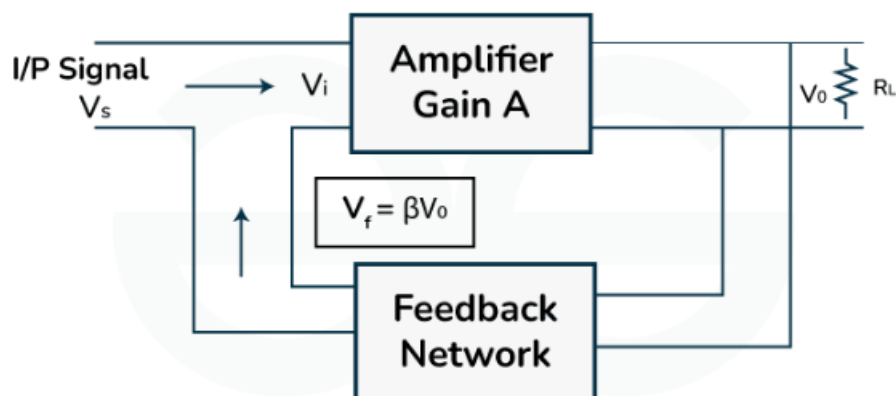
Voltage-Series (Series-Shunt) Feedback Amplifier

A Voltage-Series (Series-Shunt) Feedback Amplifier is a negative feedback amplifier in which:

The output voltage is sampled (shunt connection at the output).

A fraction of this voltage is fed back in series with the input signal.

Block Diagram



Where:

V_s = Input signal

$V_f = \beta V_o$ = Feedback voltage

V_o = Output voltage

β = Feedback factor

The effective input to the amplifier is:

$$V_i = V_s - V_f$$

The closed-loop voltage gain is:

$$A_f = \frac{A}{1 + A\beta}$$

where:

A = Open-loop gain

A_f = Closed-loop gain

Working

1. The input signal V_s is applied to the amplifier.
2. The amplifier produces an output voltage V_o.
3. A fraction (βV_o) of the output voltage is taken through the feedback network.
4. This feedback voltage is connected in series with the input and is opposite in phase (negative feedback).
5. The effective input becomes V_i = V_s – V_f.
6. If the output increases, the feedback voltage also increases, reducing the input and stabilizing the output gain.

Advantages

- Stable voltage gain.
- Reduced distortion.
- Reduced noise.
- Improved frequency response.
- Increased bandwidth.
- Better linearity.
- Less sensitive to temperature and component variations.

Effect on Input and Output Resistance

1. Input Resistance

In voltage-series feedback, the input resistance increases.

$$R_{if} = R_i(1 + A\beta)$$

where:

R_i = Input resistance without feedback

R_{if} = Input resistance with feedback

Reason: Since feedback is connected in series with the input, it opposes the input signal, making the amplifier appear to have a higher input resistance.

2. Output Resistance

The output resistance decreases.

$$R_{of} = \frac{R_o}{1 + A\beta}$$

where:

R_o = Output resistance without feedback

R_{of} = Output resistance with feedback

4. Explain the Voltage-Shunt (Shunt-Shunt) Feedback amplifier with a neat diagram. Discuss its operation and its effect on amplifier performance.

Answer:

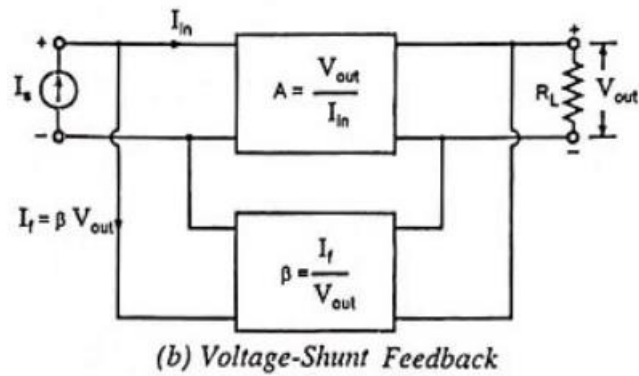
Voltage-Shunt (Shunt-Shunt) Feedback Amplifier

A Voltage-Shunt (Shunt-Shunt) Feedback Amplifier is a negative feedback amplifier in which:

The output voltage is sampled (shunt connection at the output).

A fraction of the output voltage is fed back in parallel (shunt) with the input signal.

Block Diagram



Where:

I_s = Input current/source

V_o = Output voltage

$V_f = \beta V_o$ = Feedback voltage

β = Feedback factor

The closed-loop gain is:

$$A_f = \frac{A}{1 + A\beta}$$

where:

A = Open-loop gain

A_f = Closed-loop gain

Operation

1. The input signal is applied to the amplifier.
2. The amplifier produces an output voltage V_o .
3. A fraction of the output voltage (βV_o) is sampled from the output.
4. This feedback signal is connected in parallel (shunt) with the input.
5. The feedback opposes the input signal, reducing the effective input and stabilizing the amplifier gain.
6. Any increase in output causes a corresponding increase in feedback, which automatically reduces the input signal.

Effect on Amplifier Performance

Gain: Decreases, but becomes more stable and predictable.

Bandwidth: Increases, allowing the amplifier to operate over a wider frequency range.

Distortion: Reduces significantly due to negative feedback.

Noise: Internal amplifier noise is reduced, improving signal quality.

Frequency Response: Becomes flatter and more uniform over a wider frequency range.

Stability

Gain becomes less sensitive to changes in transistor parameters, temperature, and power supply variations.

Input Resistance

Decreases because feedback is connected in shunt with the input.

$$R_{if} = \frac{R_i}{1 + A\beta}$$

Output Resistance

Decreases because the output voltage is sampled.

$$R_{of} = \frac{R_o}{1 + A\beta}$$

Advantages

- Stable amplifier gain.
- Reduced distortion.
- Reduced noise.
- Wider bandwidth.
- Improved frequency response.
- Better linearity.
- Lower output resistance.
- Better voltage regulation.

5. Explain the Current-Series (Series-Series) Feedback amplifier with a neat diagram. Discuss its characteristics, input resistance, output resistance, and applications.

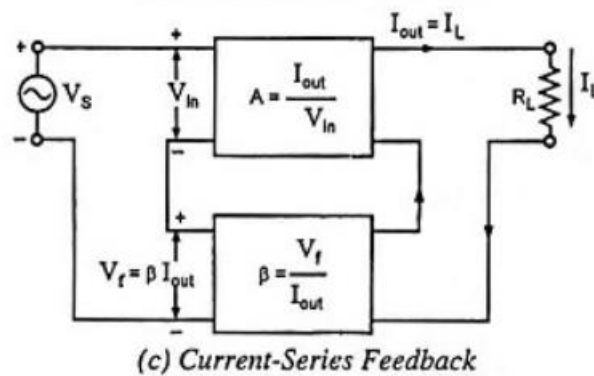
Answer:

Current-Series (Series-Series) Feedback Amplifier

A Current-Series (Series-Series) Feedback Amplifier is a negative feedback amplifier in which:

- The output current is sampled (series connection at the output).
- A portion of this current is converted into a feedback voltage and applied in series with the input signal.

Block Diagram



Where:

V_s = Input voltage

I_o = Output current

$V_f = \beta I_o$ = Feedback voltage

β = Feedback factor

The effective input is:

$$V_i = V_s - V_f$$

The closed-loop gain is:

$$A_f = \frac{A}{1 + A\beta}$$

where:

A = Open-loop gain

A_f = Closed-loop gain

Working (Operation)

1. The input signal V_s is applied to the amplifier.
2. The amplifier produces an output current I_o .
3. A portion of the output current is sensed through the feedback network.
4. This current is converted into a feedback voltage ($V_f = \beta I_o$).
5. The feedback voltage is connected in series with the input signal.
6. Since the feedback is negative, it opposes the input, stabilizing the amplifier gain and improving performance.

Characteristics

- Stable gain.
- Increased bandwidth.
- Reduced distortion.
- Reduced noise.
- Improved frequency response.
- Better linearity.
- Less affected by temperature and transistor parameter variations.

Effect on Input Resistance

Since the feedback is connected in series with the input:

$$R_{if} = R_i(1 + A\beta)$$

- Input resistance increases.
- A higher input resistance causes less loading on the signal source.

Effect on Output Resistance

Since the output is sampled as current (series sampling):

$$R_{of} = R_o(1 + A\beta)$$

- Output resistance increases.
- High output resistance makes the amplifier behave more like an ideal current source.

Advantages

- Stable gain.
- High input resistance.
- High output resistance.
- Lower distortion.
- Reduced noise.
- Improved bandwidth.
- Better linearity.

Applications

- Current amplifiers.
- Transconductance amplifiers.
- Constant-current source circuits.
- Sensor signal conditioning circuits.
- Industrial instrumentation.
- Current-controlled control systems.

6. Explain the Current-Shunt (Shunt-Series) Feedback amplifier with a neat diagram. Write Expressions for input resistance, output resistance, and applications.

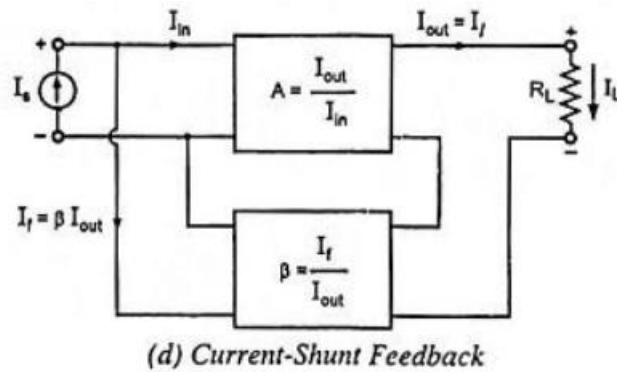
Answer:

Current-Shunt (Shunt-Series) Feedback Amplifier

A Current-Shunt (Shunt-Series) Feedback Amplifier is a negative feedback amplifier in which:

- The output current is sampled (series connection at the output).
- A portion of this current is fed back in parallel (shunt) with the input.

Neat Block Diagram



Where:

I_s = Input current

I_o = Output current

$I_f = \beta I_o$ = Feedback current

β = Feedback factor

The effective input current is:

$$I_i = I_s - I_f$$

The closed-loop gain is:

$$A_f = \frac{A}{1 + A\beta}$$

where:

A = Open-loop gain

A_f = Closed-loop gain

Working (Operation)

1. The input current is applied to the amplifier.
2. The amplifier produces an output current I_o .

3. A fraction of the output current is sampled through the feedback network.
4. This feedback current ($I_f = \beta I_o$) is connected in parallel (shunt) with the input.
5. Since the feedback is negative, it opposes the input current, reducing the effective input current.
6. The amplifier gain becomes stable, and overall performance is improved.

Characteristics

- Stable gain.
- Increased bandwidth.
- Reduced distortion.
- Reduced noise.
- Improved frequency response.
- Better linearity.
- Less sensitive to temperature and component variations.

Expressions for Input and Output Resistance

1. Input Resistance

Since the feedback is connected in shunt at the input:

$$R_{if} = \frac{R_i}{1 + A\beta}$$

where:

- R_i = Input resistance without feedback
- R_{if} = Input resistance with feedback

Effect: Input resistance decreases.

2. Output Resistance

Since the output current is sampled (series sampling):

$$R_{of} = R_o(1 + A\beta)$$

where:

R_o = Output resistance without feedback

R_{of} = Output resistance with feedback

Effect: Output resistance increases.

Advantages

- Stable gain.
- Reduced distortion.
- Reduced noise.
- Improved bandwidth.
- Better frequency response.
- High output resistance.
- Better current amplification.

Applications

- Current amplifiers.
- Current-controlled circuits.
- Constant-current source circuits.
- Instrumentation amplifiers.
- Sensor interface circuits.
- Industrial measurement and control systems.

SHORT ANSWER QUESTIONS

1. What is an oscillator?**Answer:**

An oscillator is an electronic circuit that converts DC power into a continuous AC signal without requiring an external input signal. It produces periodic waveforms such as sine, square, or triangular waves.

2. State the conditions for sustained oscillations (Barkhausen Criterion).**Answer:**

The two conditions for sustained oscillations are:

1. The loop gain must be equal to unity $A\beta = 1$
2. The total phase shift around the feedback loop must be 0° or 360° (or an integral multiple of 360°).

3. Why is positive feedback required in an oscillator?**Answer:**

Positive feedback returns a part of the output signal in phase with the input, compensating for circuit losses and maintaining continuous oscillations.

4. What is the frequency of oscillation of an RC Phase Shift Oscillator?**Answer:**

The frequency of oscillation is

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

where:

R = Resistance (Ω)

C = Capacitance (F)

5. Write the frequency of oscillation of a Wien Bridge Oscillator.

Answer:

$$f = \frac{1}{2\pi RC}$$

where **R** and **C** are the resistance and capacitance of the Wien bridge network.

6. Write the frequency equations of Hartley and Colpitts oscillators.

Answer:

Hartley Oscillator

$$f = \frac{1}{2\pi\sqrt{(L_1 + L_2 + 2M)C}}$$

where

L_1, L_2 = Inductances

M = Mutual inductance

C = Capacitance

Colpitts Oscillator

$$f = \frac{1}{2\pi\sqrt{LC_T}}$$

where

$$C_T = \frac{C_1 C_2}{C_1 + C_2}$$

Essay Questions with Answers :-

1. Explain the conditions for oscillations. Discuss Barkhausen Criterion.

Answer

Definition

An oscillator is an electronic circuit that generates AC signals from a DC supply without any external input signal.

Conditions for Oscillation (Barkhausen Criterion)

An oscillator will produce sustained oscillations if the following conditions are satisfied:

1. Loop Gain Condition

$$A\beta = 1$$

where

A = Amplifier gain

β = Feedback factor

If

$A\beta < 1$: Oscillations die out.

$A\beta = 1$: Sustained oscillations occur.

$A\beta > 1$: Oscillation amplitude increases until limited by circuit nonlinearity.

2. Phase Condition

The total phase shift around the loop must be

$$0^\circ \text{ or } 360^\circ$$

Thus, the feedback signal reinforces the input.

Importance

1. Determines whether oscillations start.
2. Ensures constant amplitude.
3. Used in designing all oscillator circuits.

2. Explain the RC Phase Shift Oscillator with circuit, working, and frequency of oscillation.

Answer

Principle

An RC Phase Shift Oscillator uses three RC sections to provide 180° phase shift, while the transistor amplifier provides another 180° , resulting in a total phase shift of 360° .

Construction

- Transistor amplifier
- Three RC sections
- Positive feedback network

Working

1. Noise initiates oscillations.
2. RC network produces 180° phase shift.
3. Amplifier contributes another 180°.
4. Positive feedback sustains oscillations.

Frequency

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

Gain Requirement

Minimum amplifier gain

$$A = 29$$

Advantages

- Simple circuit
- Low cost
- Good sine-wave output

Disadvantages

- Limited to audio frequencies
- Low output power

Applications

- Audio-frequency generators
- Signal generators
- Laboratory instruments

3. Explain the Wien Bridge Oscillator with circuit, working, and frequency expression.

Answer

Principle

The Wien Bridge Oscillator uses a Wien bridge RC network to generate stable sine-wave oscillations.

Construction

- a) Operational amplifier
- b) Lead-lag RC bridge
- c) Positive feedback network
- d) Automatic gain control (lamp or diodes)

Working

- The bridge network provides zero phase shift at one frequency.
- Positive feedback satisfies Barkhausen criterion.
- Gain control stabilizes amplitude.

Frequency

$$f = \frac{1}{2\pi RC}$$

Condition for Oscillation

Amplifier gain

$$A = 3$$

Advantages

- Very low distortion
- Excellent frequency stability
- Wide tuning range

Applications

- Audio oscillators
- Function generators
- Test equipment

4. Explain the Hartley Oscillator with circuit, working, frequency expression, advantages, disadvantages, and applications.

Answer

Principle

Hartley oscillator is an LC oscillator in which feedback is obtained through a tapped inductor.

Construction

- a) Transistor amplifier
- b) Two inductors L_1 and L_2
- c) One capacitor C

Working

- DC supply energizes the circuit.
- LC tank circuit stores energy.
- Energy oscillates between inductor and capacitor.
- Feedback from tapped coil maintains oscillations.

Frequency

$$f = \frac{1}{2\pi\sqrt{(L_1 + L_2 + 2M)C}}$$

Advantages

- Simple construction
- Easy tuning
- Suitable for RF generation

Disadvantages

- Frequency affected by mutual inductance
- Lower stability than crystal oscillator

Applications

- RF transmitters
- Radio receivers

- Signal generators

5. Explain the Colpitts Oscillator with circuit, working, frequency expression, advantages, disadvantages, and applications.

Answer

Principle

The Colpitts oscillator uses a capacitive voltage divider for feedback.

Construction

- Transistor amplifier
- One inductor
- Two capacitors

Working

- Energy oscillates between inductor and capacitors.
- Capacitive divider provides positive feedback.
- Oscillations continue as Barkhausen criterion is satisfied.

Frequency

Equivalent capacitance

$$C_T = \frac{C_1 C_2}{C_1 + C_2}$$

Frequency

$$f = \frac{1}{2\pi\sqrt{LC_T}}$$

Advantages

- Better frequency stability
- Suitable for high frequencies
- Less effect of stray capacitance

Disadvantages

- Circuit tuning is comparatively difficult

- Requires matched capacitors

Applications

- RF oscillators
- Communication transmitters
- Signal generators